Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 — Unit 3 · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26MA141A · Unit_03: Unit 3
2. 2 · Formation of PDE
3. 3 · Introduction
4. 4 · Problems on formation of PDE by eliminating arbitrary functions
5. 5 · Session-3
6. 6 · Lagranges Linear equation_continued
7. 7 · Lagranges Linear equation_continued
8. 8 · Method of separation of variables
9. 9 · Modeling of One dimensional Heat equation
10. 10 · Solution of one dimensional heat equation
11. 11 · Higher order PDE
12. 12 · Particular integral of type 1 and 2
13. 13 · Problems on finding the solution when RHS=0
14. 14 · Particular integral -Type 3 and Type 4
15. 15 · Source slide deck: Lagrange s Linear Equation
16. 16 · Source slide deck: Problems of Lagranges equation
17. 17 · Source slide deck: Method of separation of variables continued
18. 18 · Source slide deck: Particular integral of Type 1 and 2 Continued
19. 19 · Source slide deck: Particular integral of type 3 and 4 continued
20. 20 · Source slide deck: Particular integral when X is any other form
21. 21 · Source slide deck: Particular integral when X is any other form continued
22. Appendix A · Symbol glossary
23. Appendix B · Sources, coverage and gaps
24. Appendix C · Revision and retrieval practice

M1

Unit_03 · Unit 3

1 UE26MA141A · Unit_03: Unit 3

This unit master organizes source material for Engineering Mathematics - I, Unit_03, officially titled Unit 3. It includes 13 video transcripts and 18 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26MA141A · Unit_03: Unit 3

Official catalogue entry: UE26MA141A, Unit_03 (Unit 3). Bucket folder present: yes. Videos: 13; slide decks: 18.

EV0000001

2 Formation of PDE

Study focus: Formation of PDE. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will discuss formation of partial differential equation by elimination of arbitrary functions, solutions of partial differential equations, and solution of first order equations.

Key details

- Remember, there are two ways to form a PDE.
- That is by elimination of arbitrary constants and by elimination of arbitrary functions.
- In a similar manner, a second order equation may arise from an expression containing.
- Two arbitrary functions.

- Remember, it is not always guaranteed that an nth order PDE can be formed from an equation involving n arbitrary functions.
- In some cases, higher order.
- Partial derivatives may be required to eliminate all n arbitrary functions.
- That leads to a PDE of harder higher than n.
- In these cases, the obtain PDE may not be unique.
- For a partial differential equation from an arbitrary function z equals f of x plus y.
- Observe that in this question we have f to be an arbitrary function.
- Because there is only one arbitrary function.
- We proceed by finding first order partial derivatives.
- That's going to be f dash of u times dou by dou x.
- Since u equals x plus y, dou u by dou x becomes 1.
- Because U is a function of X and T.
- So w by dou x becomes f dash of x plus ct plus g dash of x minus c t and w by dou t equals f dash of x plus c t times c plus g dash of x minus ct times minus c because the coefficient of minus c t with respect to t is minus c.
- Since there are two arbitrary functions, we need second derivative also.
- We get f dash of x plus ct plus g double dash of x minus ct.

Example or application from the lecture

- If a given equation contains a single arbitrary function, we can always form a first order PDE.
- Let's take up some of the examples.
- Then the given function becomes z equals f of u.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

In dou square u by dou t square, if you take c square common, we are left out with.

Key details

- F double dash X plus C T plus G double dash X minus C T which is same as dou square u by dou x square hence the second order PDE will be dou square u by dou t square equals.
- C square times dou square u by dou x square.
- Observe carefully in this question we have a single arbitrary function, so we can expect first order PDE and maybe more.
- Differentiate partially with respect to x.
- Observe dou z by doux and double dou z by dou y.
- So eliminate f dash by dividing the first equation by the second.
- From the PDU for this particular function.
- Since z is a function of x and y, we can find out douz by doux and double dou z by dou y.
- That's same as f dash of x square plus y square times 2x plus 1 and dou z by dou y equals f dash of x square plus y square times 2y.

- We cannot solve like previous problem.
- Substitute f dash of x square plus y square in the first expression.
- That leads to dou z by dou x equals 2x of 1 by 2y dou z by dou y plus 1.
- Hence the required PDE will be dou z by dou x minus x by y times dou z by dou y equals one so remember there are two ways to form a pd first one by elimination of arbitrary constants.
- And second by elimination of arbitrary functions.
- Remember, a solution of a PDE is nothing but.
- It is known that a PD can be formed in two ways, one by eliminating arbitrary constants, and second one by eliminating arbitrary functions.
- And these relations are solutions of partial differential equation which is formed.
- There are two types of solutions for partial differential equation.
- Solution containing arbitrary functions and solution containing arbitrary concepts.

Example or application from the lecture

- Let's look at the third problem.
- Given z equals f of x square plus y, where f is an arbitrary function.
- So finally the first order PDE after eliminating f from the given function will be.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

Let's look at what is a complete solution and what is general solution.

Key details

- A solution of a PDE is said to be complete solution if the number of arbitrary consents is same as the number of independent variables present in a PDE.
- Let Z equals f of x plus y.
- We need to form a PDE by eliminating this arbitrary function f.
- It is very easy to see that we can eliminate f dash of x space y by equating dou z by doux and dou z by dou y.
- The equation dou z by dou x minus dou z by dou y equals zero is a partial differential equation formed by eliminating arbitrary function from the original function.
- So an equation containing XYZPQ defines a first order PDE which we have seen already in the session number one.
- A relation of the form phi of UV equals zero where phi is an arbitrary function of u and v.
- and satisfies equation one is called a general solution.
- In a similar manner, the solution obtained by determining the arbitrary function in the general solution by using some specified condition is called a particular solution.
- We will look at the idea of general solution once again in Lagrange's linear equation.
- We understood the idea of solution of PDEs.
- We also learned solution of first order equations.

Example or application from the lecture

- In order to understand solution of a PDE, let's look at an example.
- As we know from previous idea, let us find out douz by doux and dou z by dou y.
- So how do we talk about solution of a PDE through this example?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

3 Introduction

Study focus: Introduction. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000014 · EV0000015

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we'll understand partial differential equations.

Key details

- Introduction to partial differential equations.
- An equation involving independent variable, dependent variable and derivatives of the dependent variable with respect to independent variable is called a differential equation.
- In this case x is the independent variable, y is dependent variable and the derivative of y with respect to x is nothing but dy by dx.
- Differential equations are classified into two based on the number of independent variables present in it.
- The first one ordinary differential equations and partial differential equations.
- In the case of ordinary differential equations, we see that the derivatives are with respect to only one independent variable.
- Also, the unknown function depends on a single variable.
- And X is independent variable.
- In a similar way, in partial differential equations we see that the partial derivatives are with respect to two or more independent variables.
- The unknown function depends on multiple variables.
- In this case u is dependent variable and x and t are independent variables.
- A partial differential equation is an equation that involves partial derivatives of a function with several independent variables.
- As in the case of ordinary differential equations, we have only one independent variable.
- In the case of partial differential equations, we have two or more independent variables.
- Some of the examples of partial differential equations are one dimensional heat conduction equation, Laplask equation, and one dimensional wave equation.

- As we have seen earlier, one of the applications of ordinary differential equations is modeling.
- In a similar manner,.
- Partial differential equations help us in understanding many real world phenomena like waves, those in vibrating strings, heat flow, fluid movement, electromagnetic fields, etc.
- As in the case of ordinary differential equations, the order of a partial differential equation is the order of the.

Example or application from the lecture

- Before we understand what is a partial differential equation, let's look at what is a differential equation.
- As an example, dy by dx plus y equals e power x.
- Let's understand the classification of differential equations.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 05:00–09:59

Highest order partial derivative present in it.

Key details

- The partial derivatives.
- As you can see, some of the examples to answer the order and degree of a partial differential equation.
- Then pqr s t are denoted by dou z by doux, dou z by dou y dou square z divided by dou x square, dou square z divided by dou x dou y and dou square z divided by dou y square.
- So based on the standard notations, an equation containing X, Y, Z, P, and Q defines a first order partial differential equation.
- Remember, this first order partial differential equation is said to be linear if it is linear in P and Q.
- It means we cannot have the product of P and Q, we cannot have P squared, we cannot have Q square, and so on.
- An equing XYZ PQRST defines a second order partial differential equation.
- That is G of XYZ PQRST equals zero.
- If it is linear in PQRST.

Example or application from the lecture

- In the first example, you can see order is also one, degree is also one.
- In the second example, order and degree are two, one, and one respectively.
- In the third example, order and degree are two and one respectively.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

4 Problems on formation of PDE by eliminating arbitrary functions

Study focus: Problems on formation of PDE by eliminating arbitrary functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000022 · EV0000023

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will solve problems on formation of partial differential equation by eliminating arbitrary functions.

Key details

- In one of the classes we understood what is the procedure to form a PDE by eliminating arbitrary functions.
- And phi becomes phi dash.
- Phi dash means phi dash of x square minus y square and x square become 2x.
- In a similar manner, differentiate z partially with respect to y.
- So we get phi as the first function.
- X plus Y becomes 1 with respect to Y and pi dash of X square minus Y square times minus 2y and this X plus Y is a another function.
- How to eliminate file.
- That is so phi dash becomes phi minus z y divided by two y of x plus y.
- It means you can take this term to right hand side left hand side, it become plus.
- x plus y times u y into phi dash.
- Then you can shift this z sub x y to right hand side so it becomes phi minus z y.
- Write x plus y and two x as it is.
- In place of this phi dash, we substitute phi minus douze by dou y over two y of x plus y.
- By the standard notation, we understood that those z by doux is same as p and this y as it is.
- So we are left out with the x by y of phi minus q because dou z by dou y equals q as per the standard notation.
- If you look at the problem.
- Say phi means phi of x square minus y square.
- This is z over x plus y.
- X by Y times Q.
- You can clearly see that's this is a first order partial differential equation.

Example or application from the lecture

- Let us note on the first problem z equals x plus y phi of x square minus y square.

- So let us take y as the first function and the derivative of x plus y with respect to x is 1 plus x plus y is a second function.
- Let us call this as expression 1.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 05:00–09:59

That's the answer.

Key details

- I hope you understood the concept.
- Z equals x power n times f of y by x.
- Even this function contains a single arbitrary function, namely f.
- We differentiate z partially with respect to x.
- So f is the first function.
- And the derivative of x power n with respect to x is nothing but n times x into x power n minus 1 plus x power n as the first function and the derivative of f of y by x becomes f dash.
- of y by x then y by x becomes minus y by x square with respect to x.
- And f becomes f dash and y by x becomes 1 by x.
- Y is 1 and x is constant.
- Again, this is going to be x power n minus two times y.
- In place of f dash, we put dou z by dou y over x power n minus one.
- So in the next step we can see this is nothing but x power n divided by x.
- And this is going to be if you use law indices for x power n minus 2 and x power n minus 1, you get x power minus 1.
- nx n f minus y into q because those that better y is q as per the standard notation so if you observe this particular step still f is same so how do we eliminate this f.
- Because here we take it is f dash of one plus dou z by dou y.
- It is nothing but this.
- When you take f dash common get one plus p.

Example or application from the lecture

- Let us look at the second problem.
- Okay in a similar manner let us compute to dou z by dou y so for douz by dou y no need to apply product rule because x becomes a constant.
- Let us substitute this into this expression.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 10:00–14:59

And this can be simplified to this particular step.

Key details

- you can just shift.
- one plus q to left hand side y z plus x y p equals one plus p time x z plus x y q so when we simplify this so that leads to this particular step so that is x of y minus z p plus.
- Y of z minus x q equals z of x minus y.
- So you decide what is the order of this linear differential equation because of presence of P and Q.
- With respect to y, we get douz by dou y equals f of x times g dash f i because f of x is constant.
- We proceed for.
- Second order partial derivatives.
- Observe carefully from either of these two.
- You can say either ZXY or it could be ZYX.
- No, this is nothing but mixed partial derivative.
- It is nothing but.
- F dash of x times g dash of y.
- Even the second is also the same.
- dou square z divided by doux dot dou y.
- As you can see, as per the standard notation.
- This is nothing but S.
- by eliminating arbitrary functions.

Example or application from the lecture

- Let us take the product of dou z by doux or dou z by dou y so that leads to this particular expression and this is same as f of x dot g of y dot f dash of x then g dash of y.
- The given problem is f of x dot g of y equals f dash of x dot g dash of y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

5 Session-3

Study focus: Session-3. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000034 · EV0000035

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will discuss Lagrange's linear equation and working rule to solve Lagrange's linear equation.

Key details

- The linear first order partial differential equation of the form capital P of XYZ times small p plus q of xyz q equals r of xyz.
- And this expression is called Lagrangius equation in two independent variables, namely x and y.
- There are three steps in the procedure.
- Then we write axillary equation in this form.
- Remember there are three ways to solve this axillary equation.
- One is by method of grouping, second by method of multipliers, and the third is the differential concept.
- We write the general solution in the form φ of u comma v equals 0.
- So remember we will solve the axillary equations by the method of grouping or method of multipliers or both to get two independent solutions, namely u equals a and u equals b where ab are arbitrary constants.
- So what is method of grouping in Lagrange's method?
- So how do we apply method of grouping for a problem?
- So once we finish writing XL equation, if we want to apply method of grouping, we will write two ratios at a time like this.
- It could be dx by capital P equals Dy by Q or Dy by capital Q equals dz by R, Dx by B equal DZ by R.
- So it is first and second ratio, or this is second and third ratio, or this is first and last ratio.
- So if integration is possible, after separating the variables, we say that method of grouping works.
- And R is that.
- It is easy to see that the variables have already been separated.
- So from the first and second, we can write.

Example or application from the lecture

- Remember the general solution of Lagrangian's linear equation is given by pi of u comma v equals zero where phi is an arbitrary function.
- And U and V are the two linearly independent solutions of the acceler equation which is given by Tx divided by capital P equals dy by capital Q equals dz by R.
- After understanding Lagrange's linear equation and its general solution, let's look at the working rule to solve Lagrange's linear equation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 05:00–09:59

dx by x equals dy by y so after integration we get log x equals log y plus log c1 so after using log m plus log v so this leads to x by y equals c2 or it would c one also.

Key details

- If we take first and third, we take x by z equal to c3.
- So this is how we can apply method of grouping for a problem.

- So as usual, when we are not able to apply method of grouping, we switch over to method of multipliers.
- So what do we do in method of multiple?
- So in method of multipliers, the task is to choose three things.
- We call them as lm.
- They are either constants or functions of XYZ.
- For that we need to look at each ratio.
- by the property of ratio proportion we can write this axle equation to be LDX plus m dy plus ndz divided by lp plus mq plus r so this is capital p so what has been done here.
- Now the condition for selection of LMN is this denominator should be zero.
- It means make sure that after choose LMN, this denominator becomes zero.
- Therefore, this can be written as this numerator LDX plus mdy plus ndz equals zero and we see that it is easy to integrate.
- And find a second solution B equals C2.
- So this is a way to apply method of multipliers.
- So remember the condition for the selection of element is going to be.
- L times capital B plus M times capital U plus N times capital R should be zero.
- Observe carefully and choose a multipliers.
- Obviously, you can observe that you can cancel y and minus y, you can cancel minus z plus z and minus six plus six.

Example or application from the lecture

- So if you take first and second and third ratios, the general solution for this problem is given by f of x by y comma x by z equals zero, or it could be phi of x by y and x by z equals zero.
- So let's look at how do we apply method of multipliers for a problem in Lagrange's equation.
- In order to understand this, let's take an example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 10:00–14:59

Only if you multiply each of these three by one.

Key details

- It means to get one independent solution the multipliers are one.
- Then each ratio becomes one into dx and one into dy and one into dz divided by one times y minus z.
- as 1 times z minus x and 1 times x minus y that gives 0.
- So this is how we apply method of multiplier for a problem.
- As I mentioned, there is one other method to find the solution in Lagrange's method.
- It's a rare case, of course.
- It is called differential concept in Lagrange's equation.
- So as usual we first write the acceler equations that is dx by capital P, Dy by capital Q and Dz by R.

- Sometimes instead of grouping or using multipliers directly, we use differential concept to form a linear combination of differential that is easier to integrate.
- Remember in this case also we have to choose multipliers.
- The multipliers may be chosen such that the numerator that is LDX plus MDY plus N T Z is an exact differential of the denominator LP plus MQ plus NR.
- Once we achieve this, we will equate this with the fraction of acceler equation that is dx by p equals dy by q and dz by r.
- plus z into dz in this case we say that the numerator is exact differential of denominator and this will be equated with a fraction of dx by capital B equals dy by capital U or Dz by R anyone so to get the answer.
- What are the three different types to solve Lagrangian's equation?
- In the next session, we will solve problems on Lagrangian's equation.

Example or application from the lecture

- For example, in the denominator you may be having x square plus y square plus z square.
- Suppose if you are able to get two times x into dx plus y into dy.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

6 Lagranges Linear equation_continued

Study focus: Lagranges Linear equation_continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000046 · EV0000047

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will solve problems on Lagrange's linear equation.

Key details

- Q equals R of XYZ.
- After the selection of PQR, we write the acceler equations.
- That is dx divided by y square z over x, dy by x and dz by y square.
- to be x times dx divided by y square z equals dy by x z and dz by y square.
- So we have to think which method is appropriate after writing such a equation.
- Should we go for grouping or should we go for multipliers or can go for both.
- When we observe these actual equations, we can solve this problem by method of grouping because we can separate the variables and we can perform integration.
- we'll get x square dx and y square dy.
- And this is same as x power 3 minus y power 3 equals 3c, which is same as some constant a.
- So the first solution for this problem is going to be x power 3 minus y power 3 equals a.

- Again, when we'll observe this exer equation, we can go for grouping again.
- So if you take a look at this, if you take first and the last ratios.
- We see that x dx divided by y square z equals dy dz divided by y square.
- After cancellation of y square and this y square, that leads to x into dx equals z time dz.
- Again variables have been separated.
- As from theory, once we find out these two linearly independent solutions, we write the general solution to be phi of u, comma v 0 so which is same as phi of x power 3 minus y power 3 and x square minus z square equals 0.
- So in this case we took the first and second fractions and first and last fractions.
- If it is possible to take second and third, it is also allowed.

Example or application from the lecture

- On noting these idea, let us solve problems on Lagrangian's linear equation.
- Let us look at yz square divided by x times small p plus x q equals y square.
- In order to make simplification easy, let us simplify the first term.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let us take a look at the second problem.

Key details

- It is yzp minus xzq equals xy.
- Dy divided by minus x and dz by xy.
- Again we can easily decide we can go for method of grouping in order to find both the solutions.
- So from the first and second fractions, we get dx by y equals dy by minus x.
- After integration that becomes x square by 2 plus y square by 2 equal to some constant c1.
- That is same as x square plus y square equal to 2 c1.
- So this is the first solution for the problem.
- So to find the second solution again we can go for the first and third fractions that gives.
- Dx by Y Z and Dz by XY After cancellation of this y and this y variables are separated.
- This is same as x into dx and z into dz.
- So this is how we can apply method of grouping for a problem to solve Lagrangius linear equation.
- This is going to be capital Q and this is going to be R.
- So while looking at the actual equation, if you go for first and two fractions after cancellation of Z square plus XY and this Z square plus XY we get dx by x equals dy by minus y.
- And you can cancel this z also.
- If you take a look at right hand side, unless we solve for this xy, we cannot say this is purely a function of z.
- In order to do that, we can use the previous answer.

- that is xy equal to c1 so when you replace x y by c1 so we get this there so x power 3 dx equals z power 3 plus c1 z time dz now you can see the left hand side we have a x and dx on the right hand side we have z and dz so now you can integrate.

Example or application from the lecture

- Let us take out another problem Z of Z square plus XY times p x minus q, everything is a product.
- So now how do you find out second solution that is b equals to b let us take out the first and third fractions so we get the x power 3 dx and z power 3 plus xy z tim z.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 10:00–14:59

This is z power 4 by 4, c1 is constant, z becomes z square by 2 plus another constant, namely c2.

Key details

- So these are the problems on Lagrangius linear equation and to find out the solution we use the method of grouping.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

7 Lagranges Linear equation_continued

Study focus: Lagranges Linear equation_continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000058 · EV0000059

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In the last in the last class, we solved problems on Lagrangius linear equation using method of grouping.

Key details

- So in this session, we'll solve problems on Lagrangius linear equation by using the method of multipliers as well as.
- As we can see this is capital P.
- And this is small p and this is q and this is small q this is r after choosing p q r the axillary equations are dx over capital p, dy over capital q and dz by r.

- It is easy to see that we can't apply the method of grouping for this case because we are not able to separate the variables.
- In order to proceed with the method of multipliers, we need to choose element.
- They are either constants or functions of xyz.
- So to find the first solution, we choose a multipliers as x, y, and z.
- Then each fraction of this expression 1 becomes x into dx plus y into dy plus z into dz.
- It means we multiply this dx by x, dy by y and dz by z.
- Even in the denominator, we multiply this capital P by X.
- capital Q by Y and R by Z.
- And it is easily integrable.
- Therefore, the first solution for this case is x square plus y square plus z square equals c1.
- In a similar manner, we choose another set of multipliers, namely LMN.
- In this case, again we get the denominator as 0.
- LMN are concept.
- Lx plus MY plus Lz equals C2.
- This is nothing but V of XYZ equals C2.
- So this is how we can apply method of multipliers for in order to solve Lagrangius linear equation.
- X of Z square minus Y square.
- This is small p.
- Y of x square minus z square, which is capital Q, this is small q, and this is r.
- Again, it is easy to see that we can't apply method of grouping, so we go for method of multipliers.
- By choosing multipliers as xyz, then each fraction of one becomes x into dx plus y into dy plus z into dz.

Example or application from the lecture

- Let us solve the problem Mz minus N y times douz by doux plus nx minus Lz times douz by dou y equals L y minus mx.
- and V of XYZ equals C2 and hence the general solution for the given problem is p of u comma v equals 0.
- Let's look at another problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 05:00–09:59

Obviously in the denominator all six term gets cancel out.

Key details

- U of x y z equals c one to be x square plus y square plus z square equals c one.
- and plus x square and minus x square.
- This is another set of multipliers for finding the second which is nothing but v of x y z equals c2.

- For convenient purpose, instead of C2 we write log because all terms on the left hand side containing log.
- then xy z becomes c2.
- It is very easy to see that to find the first solution we can go for second and third ratio.
- So on integration we get y by z equals c1 because we get log y.
- equals log z plus log c one.
- Log a minus log b is log a by b so based on that we get y by z equals c one.
- So we are going for method of multipliers.
- So in this problem we apply both method of grouping as well as method of multipliers.
- Obviously by two by using the second and third fractions or ratios.

Example or application from the lecture

- It is easier to see from the given problem if x square if x is not there, you can cancel z square, or if y is not there and z is not there, you can cancel z square and minus z square and minus y square plus y square.
- Let's take up another problem y square plus z square of p minus x y q plus zx equals 0.
- And hence the general solution is given by phi of y by z x square plus y square plus z square equals zero.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 10:00–14:59

We get dy by y and dz by z so on integration.

Key details

- Should we go for grouping again or we can go for multipliers or we should go for differential concept.
- Even multipliers doesn't work.
- As from theory, this is similar to method of multipliers.
- LP plus MQ plus Nr.
- This is equated with a fraction of this is dx by p or dy by q or dz by r in order to get the integral.
- It means once we achieve this, we're going to equate this either with the first one or second one or third one.
- Depends on the problem.
- That is differential concept.
- So it means how do we do simplification?
- plus z divided by so multiply this by x so that gives x power three minus xy square minus x z square multiply second by y so we'll get a two x y square.
- The last one by z we get 2x square.
- Again same numerator.
- In the denominator you can take x common.
- It could be first one, second one, or third one.

- So unfortunately for this problem we cannot equate this with the first fraction because it doesn't have x.
- So we can equate this either with the second one or third one because it has got a single x here.
- We should observe carefully.
- is the exact differential of denominator.
- In the denominator, as usual, we have x square plus y square plus z square.
- On the right hand side we have dz by z.
- Observe the left hand side carefully, it is like the numerator is the exact differential of denominator.
- So the integration of this gives log of x square plus y square plus z square and right hand side we get log z plus log b for convenient purpose because other two terms are log.
- Therefore, if you use log a plus log b we get the solution as x square plus y square plus z square divided by z equals b.
- So in this problem to get the first solution we use the as usual grouping method.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 15:00–19:59

To get the second solution we use the differential concept method.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

8 Method of separation of variables

Study focus: Method of separation of variables. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000074 · EV0000075

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In last two classes, we learned how to solve problems on Lagrange's linear equation by using method of multipliers or method of grouping or both.

Key details

- In some special cases, we also applied differential method.
- So in this session, we will learn what is a method of separation of variables.
- How do we apply this for a problem?
- And what is its advantage.
- As an application,.

- Separation of variables provides closed form solutions for many fundamental partial differential equations, namely one dimensional heat equation, the wave equation, and the Laplace equation.
- By reducing each PDE to a set of ordinary differential equations.
- In a future class, we will see how to apply this method of separation of variables to find out the various possible solutions of one dimensional heat equation.
- Before that, we'll understand.
- How to apply this method for a problem.
- U equals f of x y as usual u is dependent and x and y are independent variables.
- Then the partial derivatives present in the equation should be.
- reduced to ordinary derivatives.
- Assuming u is a function of x and y.
- ordinary partial derivative into ordinary derivative in a similar manner if we have dou u by dou y.
- Because y is x y, this can be written as in this case we take x outside and written as d capital y divided by d small y.
- In terms of ordinary derivative.
- Or the partial derivatives are reduced to ordinary derivative in this manner.
- In other words.
- The expression containing ordinary derivatives should be rearranged in such a way that left hand side is a function of x only and rhs is a function of y only.
- That's what you can see in expression number two.

Example or application from the lecture

- This method is also called product method because we assume the solution of the given PDE in the form of a product is depending on only one independent variable.
- Let us assume that.
- If the given PD contains x and y to be independent variables, we need to assume the solution of the given PDE in the form of a product that is u equals x of x times y of y where x is a function of x only and this is a function of y only.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 05:00–09:59

After we have this scenario that is f of x equals g of y, so mathematically they both have some common constant k.

Key details

- Therefore, we equate both sides to some common constant k.
- Remember, this is nothing but ordinary derivatives.
- We need to solve these two by known methods.
- We solve these two for capital X and capital Y.
- It means the one what you got from step number two.

- So this is how we apply method of grouping, sorry, method of separation of variables for a problem.
- Use the method of separation of variables to solve this partial differential equation, which is nothing but three times dou u by doux plus two times.
- dou u by dou y equals zero so conditions are u of x comma zero equals four of e power minus x so it is not necessary to have conditions like this we can apply this method for problems without this condition also.
- That is u equal to x of x times y of y.
- So it means as we know from the procedure.
- This three dou by dou x becomes three capital y e capital X divided by D small X plus second term will be two times X into P y by.
- P y equals zero.
- This is what we assume.
- Then dx by x.
- We just ship this right hand side.
- Then we get dy by y.
- It means we write one by x by t x equals k.
- Another partition minus two by three.
- One by capital Y by DY equals k.

Example or application from the lecture

- Let us say that f of x equals k as well as g of y equals k.
- In order to understand this procedure, let's look at one or two examples.
- So let's see how do we use this condition in later stage?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 10:00–14:59

K1 constant for convenient purpose.

Key details

- Then X becomes C1 E power KX where C1 becomes e power K1.
- So this is the first solution.
- After integration it is log y equals minus 3 by 2 k y plus some constant k2.
- Therefore y equals c2 times e power minus 3 by 2 times y.
- U equals XY.
- E power A times E power B minus B, E power A minus B.
- You can take K.
- outside so that is same as x minus three by two of y so in the problem if you don't have any conditions like this.
- we can say we can stop here.
- So whenever we have a condition we need to proceed.
- Obviously, we need to find out a value as well as this k value.

- So when we apply the condition, if you observe here 4e power minus 6 equal to u of x comma 0, it means wherever we have y, we put 0.
- I mean if you substitute both A and C K value.
- We get this.
- I think now you understood the meaning of this statement in the pro in the data.
- It means by reducing each PDE to a set of ordinary differential equations as a each depending on only one independent variable.
- So it means u of x dot x dot y.
- And this is how we can convert partial derivative into ordinary derivative.
- And the condition is u of x.
- comma zero equals three times e power two x.
- So to eliminate y in the first and x in the second, we should divide through by the assumption, which is xy.

Example or application from the lecture

- That is u of x y to be the this is a solution for the given problem.
- Seven is substitute both in the given problem.
- So finally this is a solution for the given problem by using method of separation of variables.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 15:00–19:59

four by y by.

Key details

- It is not compulsory.
- Therefore x is some a1 k.
- So this is the first solution.
- or this is a solution of this ordinary differential equation that is 1 by x dx by x.
- In a similar manner how do we solve second one?
- Software solving for Y.
- So when we use a condition.
- We are equating this part to three times e power two x equals c as it is.
- Of course in our case, okay, you can call some c e power kx.
- When y is zero, so this part disappears.
- it becomes u equals three e power this is actually two of two of x plus pi by four times y.
- So therefore three power this is two x plus five by two y.
- So in this session we understood what is the method of separation of variables, what happens, and we also learned how to solve problems using this method.
- So in the next class, we apply this method to find out the various possible solutions of one dimensional heat equation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

9 Modeling of One dimensional Heat equation

Study focus: Modeling of One dimensional Heat equation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will discuss the applications of partial differential equations.

Key details

- And this alpha square is thermal diffusivity.
- It measures the rate at which.
- Heat spreads through a material.
- Observe the diagram.
- The rod is made of one uniform material that is.
- Every part of the rock is made of exact the same substance so that its properties do not change from place to place.
- Let ρ be the density of the material, c be the specific heat of the material and k be the thermal conductivity of the material.
- We have an assumption here.
- That is we assume that the rod's outer surface is perfectly insulated.
- It means heat can only travel along the length of the rod.
- Let u of x comma t be the temperature at position x and at time t.
- The quantity of heat present in the slab.

Example or application from the lecture

- Heat equation is a fundamental partial differential equation in mathematics and physics that describes the distribution of heat in a given region over time.
- Its mathematical formulation is given by dou u by dou t equals alpha square dou square u divided by dou x square.
- Consider a long thin metal rod that has same cross section area that is A along its length.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then the quantity of heat present in the slab is going to be the product of specific heat, how much so it can store, and the mass of the element slab, and the change in temperature.

Key details

- Initially, we assume that C with the specific heat and.
- Change in temperature obviously delta u.
- And this volume is nothing but a times delta x.
- This is what we have seen in the diagram.
- The rate of change of heat in the slab or slice at time t.
- Let R1 be the rate of inflow of heat at X and R2 be the rate of outflow of heat at del X plus delta X.
- So it means.
- We get R C times A rho delta x dou by dou t equals R1 minus R2.
- In one dimension, Fourier's law says that the heat flow, heat flux.
- that is heat per unit area per unit time is directly proportional to the negative of the temperature gradient so we get minus k times dou u by dou x at x so.
- When we multiply the flux by cross-sectional area that is capital A, we are going to get the total rate of heat flow through the phase.
- So and this negative sign is because heat flows from a higher temperature to the lower temperature.
- When you start moving from left to right, it is easy to see that temperature decreases.
- And this is called the thermal diffusivity of the material.
- those value divided by dou x square so where alpha square equals k by 0 is the thermal diffusivity of the material so it talks about heat spreads through a material so on so in this session.

Example or application from the lecture

- Let us discuss R1 and R2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 10:00–14:59

We understood how to derive the expression for dou u by dou t equals dou square alpha square times dou square u by dou x square as a model.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

10 Solution of one dimensional heat equation

Study focus: Solution of one dimensional heat equation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000102 · EV0000103

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In previous classes, we understood what is the method of separation of variables.

Key details

- Why did we use method of separation of variables?
- We also solved some of the problems using method of separation of variables.
- In this session also, we will find the various possible solutions of one dimensional heat equation by using method of separation of variables.
- It means x and t are independent variables.
- In the form of a product that is u equal to x t where x is a function of x only and t is a function of t only.
- Then C square the right hand side term can be written as t of d square x divided by dx square.
- If you look at the LH system, we need to remove this X.
- If you look at the right hand side, we need we need to remove this T.
- C square by capital X, T square X divided by P x square.
- This can also be written as 1 by C square t dt by dt equals 1 over x d square x divided by dx square.
- We have expressed the left hand side as a function of P and the right hand side as a function of X.
- And similarly 1 by c square t.
- D by D T equals.
- of x equals zero where capital D stands for D by DX.
- In a similar manner.

Example or application from the lecture

- As in Open Theory, the one the dimensional heat equation is given by dou u by dou t equals c square times dou square u divided by dou x square.
- Then we need to substitute this assumption into the given.
- After we substitute u equals x t into the given problem, we need to convert the partial derivatives present in this equation as the ordinary derivatives so this is nothing but the left hand height left hand side can be written as x of dt by dt.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 05:00–09:59

C square capital T when you rearrange this is written as T minus K C square of E equals zero.

Key details

- Mathematically, K is an arbitrary real constant, but different signs of k gives different ordinary differential equations in X and D.
- When you write axelor equation that gives m square equals zero.
- It means m equals zero comma zero two times.
- In a similar manner, this second expression becomes.
- This is M as D of P equals zero.
- What is the solution of these ordinary differential equations?
- If you observe X is a dependent and capital T is dependent, so we have to express capital X in terms of small X.
- As from second chapter, when the roots are repeated.
- Therefore, what is the general solution for the PD?
- In the later stage, we have to decide whether we accept this or not.
- We'll discuss this after the completion of the problem.
- That is when k equal to k n k is p square.
- then this differential equation becomes d square x divided by e x square minus p square x zero.
- X equals 0.
- This is same as t minus.
- P squared C square of P equals zero.
- Therefore accelerate equation becomes M equals sorry M minus P squared C square zero.

Example or application from the lecture

- Case number one, when k is zero, case number two, when k is positive, let us say p square, k is negative, let us say k equal to minus p square.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 10:00–14:59

Again, how do we write the general solution when the roots are real or distinct?

Key details

- So for M equal to plus R minus P, the general solution will be okay now for convenient purpose, because we have used C1, C2, C3, we will now use C1 dash, C2 dash C3 dash.
- U equals xp.
- Then e power.
- P square C square T.
- Again, we need to discuss after the end of the problem.
- Should we accept this or should we reject this?

- When k is negative, say minus b square.
- As you can see, when we proceed based on case two, we get this.
- I mean to say when k is negative that is k is minus p square then the ordinary differential equations are going to be.
- You see the roots are complex.
- sorry the first root will be m plus p square c square zero so this is only m equals minus p square c square so this is the root.
- We write the general solution to be this as well as this.
- When K is zero, K is positive, and K is negative.
- And that's what we see in expression number 567.
- This is when a case positive C one dash, C two dash and C three dash.
- These are the various possible solutions of one dimensional heat equation by using method of separation of variables.

Example or application from the lecture

- Therefore, the solution of the PDE is given by.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 15:00–19:59

Which solution is considered?

Key details

- In the sense Which solution is chosen, which solution is rejected.
- If you look at this seventh one, the seventh one is independent of the.
- In other words, since u equals c1x plus c2 times c3 is independent of t, this is rejected.
- It means expression number seven, expression seven is not considered.
- What about equation number five?
- Again you can see here as the time t increases this also increases it means as p increases u also increases therefore even the fifth one is also rejected.
- And observe E power minus C square P square t.
- It is easy to see that e power minus C square P square t decays to zero.
- As t tends to infinity.
- U decreases as the time t increases.
- That's what we see.
- Therefore, out of these three, the sixth one is considered.
- Of these three solutions, we have to choose that solution which is consistent with the physical nature of the problem.
- Since u decreases as time decreases, the only suitable solution of the 8th equation is the sixth one.
- So in this session we found out the various possible solutions of one dimensional heat equation by using the method of separation of variables.
- We also understood out of these three solutions, which one is suitable.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

11 Higher order PDE

Study focus: Higher order PDE. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000118 · EV0000119

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will discuss the solution of higher order linear partial differential equations with constant coefficients.

Key details

- In second chapter, we understood how to solve homogeneous and non-homogeneous higher order linear ordinary differential equity constant coefficients.
- Those methods can be extended to solve higher order linear partial differential equations with constant coefficients.
- That is aught times dou nz divided by dou x n plus.
- A1 times dou n z divided by dou x n minus 1 dot dou y and so on.
- And if you see on the right hand side we have f of x y.
- Power R of Z equals f of x y.
- Remember, from now onwards we use these notations.
- Small p is nothing but douz by dou x, which is same as dz, q same as douz by dou y.
- This is nothing but d dash up z.
- And this is r value.
- And this is S value.
- And similarly we have a T to be dou square Z divided by dou y square.
- This is nothing but D dash square of Z.
- We will now understand what is the general solution of this expression number two.
- Similar to the ordinary differential equations, the general solution of equal can be written as Z equals complementary function that is C dot F plus particular integral PI Where complementary function is nothing but the solution of.
- Homogeneous equation that is f of d comma d dash z equals zero, that is left hand side, and the particular integral is a solution satisfying.
- A non-homogeneous equity that is two and it doesn't contain any arbitrary constants.
- Before we understand how to compute complementary function, we need to know there is something called reducible factor and irreducible factor.
- Note that if f of d, d dash can be factorized into linear factor in the form AID plus VID dash plus c i, then this f of b, d dash is said to be reducible.

Example or application from the lecture

- Let's look at the general linear partial differential equation of order n in two variables.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let's look at d square minus a d dash square.

Key details

- As, you can use a square minus b square formula.
- This exactly looks like.
- AID plus BID dash and of course C0 it is also acceptable.
- D minus 2d dash and d minus 3d dash up zero.
- Therefore we say that this f of d comma d dash up z 0 is reducible.
- You can clearly see that this is not reducible because you can't express this in the form of EID plus B ID dash plus CI.
- Even the third one is also.
- not reducible factor because you can't express this in terms of AID plus BID dash plus ci.
- D power three minus d square d dash plus d dash minus d dash square can be written as.
- D minus d dash times d square plus d dash square up z zero.
- It is easy to see that from the first two we can take d square common and you get d minus d dash.
- From third and fourth term you can take a d dash common, you will get d minus d dash.
- The first two are reducible factors or reducible forms and equation six and seven are not reducible or irreducible forms.
- So that's the next question.
- Category number one, we may have distinct rules.
- Second category, we may have repeated rules or factors.
- Of course we call it as existing factors and repeated factors.
- This form that is aid plus bid dash plus c i of z zero.
- of BIX minus a i y.
- So if C i is zero, obviously the first term becomes one.
- So as a first case, if C i is zero, this is how we write the solution.
- In a similar manner, if a i is zero, we write the solution in this form.

Example or application from the lecture

- If you look at the second example, this can be written as D minus D dash.
- And what about the third example?
- What about the fifth example?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 10:00–14:59

In a similar manner, if Bi is 0, this is how we write the solution.

Key details

- That is e power minus c1x by a1.
- times phi 1 p1x minus a1y and so on.
- It depends on number of factors present in the problem.
- A I D plus B I D dash plus CI.
- It is easy to see that a1 is 1, B1 is minus 1.
- For the second factor, A2 is 1 and B2 is 1.
- Therefore, we can use equation 11.
- So based on this idea, so we first get z equals phi one.
- of if we multiply p1 by x we get minus x then minus y.
- We write it as some phi one star of minus x minus y.
- If we take minus common, then that is written as phi one of x plus y.
- And there's no change in the second term that is phi two of x minus y.
- And we can call this a general solution because if you look at the problem.
- The order of the PD is 2, and the number of arbitrary functions are also 2.
- So therefore we can call this as general solution.
- Plus 11 D dash square minus 6 d dash 3 of Z 0.
- So remember there is a shortcut method.
- So that shortcut method is applicable only when we have scenario like this.
- If you observe.
- The total degree of each term is the same.
- If you look at this power value, this is 3 and 2 plus 1 is 3, and 1 plus 2 is 3, and this is 3.
- So it means so we can divide this expression by.
- As D dash three.
- So this is same as divide through by d dash power 3 so the first number is d by d dash power three then second number term is going to be b by d dash square.
- Then the third term is 11 d by d dash and the last one is six so z equals zero.
- p by d dash equals m then that equation becomes m power three minus six m square plus eleven m minus six zero and you can see this is a cubic equation in m.

Example or application from the lecture

- Let us take up these four examples.
- If you look at d square minus d dash square of z zero, as we know from the example, this can be written as d plus d dash and d minus d dash of z zero.
- So let's look at the second example that is d power tree minus 6 d square dash 10 d dash.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 15:00–19:59

For example, when we write d by d dash equals 1.

Key details

- we get d equal to d dash so it is same as d minus d dash so one factor is d minus d dash similarly when m is 2 we get d by d dash equals 2 so that is same as d minus 2 d dash so this is second factor.
- At last you can write d by d dash equals 3.
- And you can see all these three are reducible factors.
- We write the solution in this form.
- And very well when the first two terms are negative, we just introduce a star, then we can take minus common that is same as phi one of x plus y, phi two of two x plus y and phi three of the x plus y.
- If all terms in f of d gamma d dash have the same total degree, then we can set d by d dash equals m and then we find the factors of the equation using m.
- This shortcut method cannot be applicable for the third and fourth examples.
- In third and fourth, the total degree of each term is not same.
- So we cannot apply shortcut method.
- Then how do we solve?
- So the above mentioned note doesn't applicable to the following two problems because the total degree of each term is not same.
- Then how do you find out the factors?
- So that should be done by a normal inspection method.
- And the total degree is 2, this is 2, and third is 2, and this is 1, this is 1, this is 0.
- 7 d dash can be written as D six d dash plus 1d dash.
- There's no change in two d dash square.
- Again this 7d can be written as 6d plus d.
- Again this 4D dash is same as two d dash plus two d dash plus two of z right so if you observe there are nine terms.
- So this can be further written as if we take D common, you will get 3D plus e dash plus one, it means you're combining this, this, and this.
- And again, if you can take two d dash common.
- Z so from here you can get this again because three D plus D dash plus one.

Example or application from the lecture

- So I hope this note is clear after looking at this second example.
- For example, in this case, we write the first term same.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 20:00–24:59

And it is easier to see that the order of the PD is 2 and there are two factors here after finding a1 b1 c 1 and a2 b2 c2 and substitutes for equation.

Key details

- we get this as a general solution e power minus x pi 3 phi 1 of x minus 3y plus e power minus 2x phi two of two x minus y.
- Here phi one and phi two are any arbitrary functions.
- That is 2d square minus d dash minus d dash square minus d plus d dash.
- Again you can observe.
- The total degree of each term is not same.
- We can split dd dash to be minus two plus one and there's no change in the other three terms.
- So this is what we get in the next step.
- If you look at odd places, you can take d common.
- If you look at e1 places, take minus t dash common.
- We get 2D plus D dash minus 1 of Z 0.
- For the first factor, a1 is 1, B1 is minus 1, C1 is 0.
- For the second factor, a2, b2, c2 are 2 1 minus 1 respectively.
- Because b1 is minus 1, we get a star that is same as phi one of x plus y.
- Then for the second because c is non-zero, we get a power x by 2 because.
- C is minus 1 minus of minus that is plus and a2 is 2, that's why x by 2, by 2 of x minus 2 y.
- when we have distinct factors or different factors.
- You have taken two for understanding theory part.
- It could be three, four, and so on.
- If BI0 we get this, if AI0 we get this.
- CI0 is a part of this.
- When we have a factor in this form, so this is how we write the resolution.
- When BI0, this is how we write the solution.
- we write the general solution in this form.

Example or application from the lecture

- Let's take another example.
- So this should be solved like the previous example.
- Suppose assume that let a factor be of multiplicity too.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

12 Particular integral of type 1 and 2

Study focus: Particular integral of type 1 and 2. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000138 · EV0000139

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will learn.

Key details

- And f of x comma y equals sine of a x plus b y as well as cos of a x plus b y.
- It means this is not zero.
- That is z equals 1 by f of d comma d dash.
- Then f of x comma y.
- So basically what we do is that we need to take up a and b from here and then replace b by a and d dash by b so the particular integral becomes one over f of a comma b tim a power a x plus b y remember we have a condition here f of a comma b.
- Should be non-zero.
- Otherwise that leads to one by zero case.
- That is Z is written as.
- Some foyer of x comma y times e power a plus b y.
- Where this faya of x comma y is nothing but f of b plus a and d dash plus b.
- It means we have d and d dash.
- Once we write foyer of x comma y to be f of b plus a d dash plus b we need to expand this as an infinite series.
- We substitute back to Z equals phi of x comma y times e power a x plus b y.
- D inverse is nothing but integral with respect to x, keeping y constant, and d dash inverse is nothing but integral with respect to y, keeping x constant.
- Find the general solution for the partial differential equation 2d square minus dd dash minus d dash square plus d.

Example or application from the lecture

- How to find the particular integral for the given partial differential equation where the right hand side is non-zero.
- As we know that the general form of the non-homogeneous partial differential equation is given by f of b, d dash of z equals f of x comma y.
- In this case, the formula for particular integral is given by z equals 1 over f of d comma d dash times f of x gamma y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 05:00–09:59

Minus D dash equals B power two x plus three y.

Key details

- It is easy to see that a values two and b values three.
- differently partially different equation can be written as 2D plus 1 and D minus D dash.

- So for the first factor, we can see this is A1 value.
- B1 is 1 and C1 is 1.
- For the second factor, A2 is 1 b2 is minus 1 and c2 is 0.
- And we know that in the divine problem a values two and b value is three, when you substitute b s two and d dash to be three.
- [ASR low confidence; check audio] e power 2x plus 3y.
- 4 d square plus 3 d dash minus d dash square minus d minus d dash of z equals this.
- As you can see a value is one by two and b values two.
- And it is easy to see that.
- When we replace d by half and d dash by two, so we get zero.
- That is phi of x comma y.
- and e power x plus two y by two.
- When you substitute this back to.

Example or application from the lecture

- Let us find out the particular integral.
- In the given problem that is f of d comma d dash, we see that this is going to be minus 8, so which is non-zero.
- And hence the particular integral for this particular case is given by minus 1 by 8 times e power 2x plus 3 by.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

Core explanation · 10:00–14:59

And we know from previous slide.

Key details

- F comma d dash to contain d square, d dash and d dash square.
- D dash by minus a b and a dash square by minus b square.
- So the same method can be used even for cost of a x plus b y.
- In case we get denominator zero, that should be solved by general case, which we will see later.
- That is general case.
- 2 times dough Z by doux square minus 3 dou square z by doux dou y plus dou square z by dou y square.
- And the very well this is same as d square and this is same as the d dash and this is sam as the dash square.
- Are the total degree same.
- We see that.
- When we replace d square by minus one square, we get minus one.
- D dash by exactly minus two, so that become plus two here, then d dash square to be minus of minus two square, so that becomes totally minus four.
- Is minus one by twelve sine of x minus two y.

Example or application from the lecture

- Then we take out a and b from the given problem, then replace d square by minus a square.
- So let's take our example.
- Find the particular integral for the given partial differential equation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

13 Problems on finding the solution when RHS=0

Study focus: Problems on finding the solution when RHS=0. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000150 · EV0000151

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we'll solve problems on finding the general solution of homogeneous linear partial differential equations with constant coefficients.

Key details

- In the last class, we understood how to factorize.
- F of D comma D dash Z equals zero into linear factor that is AID plus AI D dash C i.
- Sometimes CI may be zero or B I may be zero or AI may be zero.
- if we get to know that the total degree of each term is the same, then we can use this shortcut to make the problem simple.
- And it clearly shows that the total degree of each term is M.
- Either we can use d by d dash to be m or we may look for.
- A plus b ol square or a minus b whole square that makes the problem even simpler.
- And this is same as 2ab that is d square and d dash square.
- A plus B and A minus B.
- The coefficient of d dash is minus 1, so b1 is minus 1, and there is no c.
- Yeah, so C1 is 0.
- As z equals x of phi 1 x plus y and phi 2 of x plus y.
- This is for the first factor.
- And this is for the second factor.
- We know a logic when we have both the terms to be negative, we take we can take minus common and that is same as this.
- Before that we introduced our.
- And now we get the three of D by D dash plus one of Z.
- Minus 3m plus 1 0.
- So we need to find out what are the three roots of m because this is a cubic equation in M.

Example or application from the lecture

- Let's look at the first problem d power four minus two d square t dash square plus d dash power four of the d plus zero.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 05:00–09:59

So when we use synthetic division we see that m equals minus 1 is the first root by inspection and then we get to know that we get 4m square minus 4m plus 10.

Key details

- And this is same as 2m minus 1 whole square zero and we see m equal to 1 by 2 two times.
- and when we put m equal to half we get d by d dash equals one by two this is same as 2d equal to d dash so this is same as 2d minus d dash and this square because m equals half repeated two times from here.
- But the first.
- D plus d dash A B C R 110 respectively and for the second ABC R 2 minus 1 and 0 respectively.
- Whenever we see B is negative, we need to go for star concept, what we know from earlier.
- So for the first factor this is the solution for the second factor this is the solution.
- And you can see the order of equation is three.
- And the number of orbital functions are also three, therefore this is a general solution written as Z equals this.
- So when we use the shortcut method we see that the right equation in m is m square minus m minus x zero so m values three and minus two so obviously when we say a by d dash equals three the first factor will be d minus three d dash.
- And the second factor is d plus 2d dash.
- And these are the abc value C0.
- Then we get the solution to be phi one of 3x plus y, phi 2 of 32x minus y.
- In a similar manner if we look at the port problem, it is easy to see that and this part is nothing but a square sorry a minus b whole square.
- That is a d minus d dash square and we have another square outside.

Example or application from the lecture

- Let us substitute m as 1 minus 1 and 1 by 2 respectively in d by d dash.
- So let's first find out a i, bi and c i values.
- Let's look at the third problem that is d squared minus d dash minus six d dash square.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

14 Particular integral -Type 3 and Type 4

Study focus: Particular integral -Type 3 and Type 4. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In previous session, we learned how to find the particular integral where f of x comma y equals sine e but x plus b y as well as sine of x plus b y or cos of x plus b y in this session.

Key details

- We will take up two more cases.
- That is finding Pi when f of x comma y is a polynomial in x and y.
- That is f of x comma y equals x power m times y power n.
- Also, we look at the general case.
- Or when f of x comma y is not is not a polynomial in x and y.
- That is when f of x comma y equals x power m dot y power n or at polynomial in x and y.
- How to how do we find out particular integral?
- As that the formula for particular integral is going to be one divided by f of d, d dash of f of f of x, y.
- As we have seen in ordinary differential equation, here also we bring f of b, comma d dash to the curator so that it's going to be z equals f of b, d dash inverse.
- X power m dot by power n.
- Then we need to expand f of b, d dash inverse as an infinite series, and then operate by x power m dot by power n.
- As a special case.
- How do you find out the particular integral when we have a polynomial on the right hand side?
- Square plus y square.
- Either we can take out d square common or d square out common because m and n are equal here.
- In case we take out d square common, what happens?
- Because in the polynomial case, so it is convenient to have either one plus something or one minus something before we apply binomial series.

Example or application from the lecture

- Let's start with case number three.
- suppose f of d gamma d dash doesn't contain the constant term that is the term which is independent of d and d dash then we have two cases we expand f of d gamma d dash inverse power minus one in power of d dash inverse dot d if m is less than n or in power sub d inverse dash dot d dash if m is greater than n.
- Okay, let's take an example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

minus square and d da square which is nothing but d square of one minus d da square divided by d square so this is going to be d na square.

Key details

- And you can easily see that this is like one minus x or one minus something that something is this b inverse square dot b dash square.
- Moreover, all these things are in the denominator.
- So in the next time we need to bring it up.
- D square minus theta square inverse of x square plus y square.
- So when we use the series, so that's going to be the minus two of one minus the inverse square then d dash square of x square plus y square because m and n are equal to two so we stop the series so when we reach d square as well as d data square.
- So because the highest power of m and n is two, we stop the series at t square and d ta square.
- So that is why if we look at the formula for binomial series, these terms are not considered because anyway.
- D inverse power four and so on of this x square plus y square become zero.
- So when you write the formula we are considering only one as well as x because x is t square and d naught square and d da square.
- So it means we are getting power minus two of one plus.
- B inverse square.
- Then we have our d dash square.
- Other terms are neglected.
- Outside we have x square plus y square.
- Then we should take it inside.
- D inverse square dot d dash square of x square plus y square.
- Next step we just read an as it is.
- So d dash square means we need to find out the partial derivative of this with respect to y.
- Then what is t inverse square of 2?
- So first time this is 2x, second time this is 2x square by 2, but cancellation 2 on 2 we get x square.
- Then what is d power minus two?
- plus x square dot y square by two.
- So as we know from the earlier concept, because m and n are equal, we have two choices.
- Either we can take d square common or we can take d dash square common.
- for a PD there are chances of getting different answers.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

Core explanation · 10:00–14:59

Okay, let's take a another example.

Key details

- Find the particular integral for the partial differential equation.
- Either we can take a t outside or t dash outside.
- Assume we take D outside, so that leads to this particular step, and we get one plus x, then that series formula minus two.
- Because the highest power of x m and n or the values of m and n r equals one, it is enough to go up to d and d dash.
- two d dash and d power minus one in it these things are zeros so once we take three x plus two y inside so that leads to three x plus two y and two d dash d power minus one three x plus two y.
- As we know, this is nothing but integration of x with respect to x, integration with respect to x, keeping y constant.
- So 3x becomes 3x square by 2, so y becomes 2xy because 2y is constant and dx become x.
- Then what is d dash of this part?
- Of course partial derivative.
- Outside we have one more 2, therefore we get minus 4x.
- We need to integrate with respect to x two times.
- plus x square y.
- In case we take out the d dash common, so that leads to this particular answer.
- You can clearly see that.
- Both answers are different and it is valid also.
- If you remember when we discuss case number two, it was said that when f of minus d square minus d dash and minus d dash square become zero.
- So this is a d square because we replace d square by minus a square, so it is a d square minus a d dash and d dash square zero.
- How do you find out P PI?
- So even that case can be solved by using this method.
- Assume that f of b, d dash is reducible.
- As from the beginning of this chapter, so if we can express f of b comma d dash to be something like AI.
- D plus BI D dash plus CI.
- If F of B gamma D dash can be reduced to this form, then we say it is reducible.
- Then how do we apply the procedure here?

Example or application from the lecture

- When a form x comma y is not any of the forms as given in the above cases or.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 15:00–19:59

Remember in this case C is taken zero.

Key details

- A1P and D dash Z is nothing but douz by dou y which is same as Q so therefore we get B1 Q equals F of X com Y.
- If you remember Lagrange's equation.
- is exactly the same so what is Lagrangius linear equation capital p of x y z times all p capital q of x y z small q equals r so it we learned that this is how lagrangius linear equation looks like so here also we have the same thing.
- First and second fractions and first and third fractions.
- So from the first and second ratios, we can get when we have variable separable that is b1 dx equals a one dy.
- After integration that leads to B1 minus A1 to Y equal to C where A C is arbitrary constant.
- Remember in later stage we need to solve for y.
- Okay, when we take the first and third fractions or first and third ratios.
- We get dx by a1 equals dz by f of x comma y.
- That is a1 dz equals f of x karma b1x minus c by a1 dot bx.
- Since we are looking for a particular integral, we need to put C1 to be zero.
- F of x comma b1 x minus a1.
- This procedure is repeated based on how many factors are involved in the problem.
- Also we looked at the general.
- That is when Mf of X comma y is not e pous b y not sine of x plus b y or cos of x plus b y or not a polynomial in x and y.

Example or application from the lecture

- Let's see.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

15 Source slide deck: Lagrange s Linear Equation

Study focus: Source slide deck: Lagrange s Linear Equation. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000261 · EV0000262

SHORT-NOTE RECAP

Study notes · Source slide deck: Lagrange s Linear Equation

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MATHEMATICS - I

- Unit - 3: Partial Differential Equations
- Department of Science and Humanities

Slide 2 · 1 Lagrange's Linear Equation

- 2 Working Rule to solve Lagrange's Linear Equation

Slide 3 · Lagrange's Linear Equation

- 1 The linear first-order partial differential equation of the
- is called the Lagrange's equation in two independent variables x, y
- 2 Theorem: The general solution of the equation Pp + Qq = R is

Slide 4 · Working Rule to Solve Lagrange's Linear Equation

- 2 Solve the auxiliary equations by the method of grouping or the
- method of multipliers or both to get two independent solutions
- 3 $\phi(u, v) = 0$ is the general solution of the equation Pp + Qq = R

Slide 5 · Method of Grouping in Lagrange's Equation

- 1 The method of grouping involves pairing two of these
- ratios at a time to form ordinary differential equations
- 4 The general solution is given by $\phi(u, v) = 0$, where u and v are the

Slide 6 · Suppose the auxiliary equations are:

- Grouping dx = dy : x y
- Z dx Z dy = =⇒ln x = ln y + c1 =⇒x = c2
- Z dx Z dz = =⇒ln x = ln z + c3 =⇒x = c4

Slide 7 · Method of Multipliers in Lagrange's Equation

- 3 l, m, n are chosen in such a way that lP + mQ + nR = 0. Thus
- 4 Solve this differential equation, and the solution is u = c1
- 5 Similarly, choose another set of multipliers (l1, m1, n1) and the

Slide 8 · Suppose the auxiliary equations are:

- y −z z −x x −y
- Choose multipliers l = 1, m = 1, n = 1. Then,
- This gives dx + dy + dz = 0. On integration, it gives x + y + z = C1.

Slide 9 · Differential Concept in Lagrange's Equation

- 1 Given the auxiliary equations: dx = dy = dz P Q R
- we use the differential concept to form a linear combination of
- 5 Finally, ldx+mdy+ndz is combined with a fraction of dx = dy = dz to lP+mQ+nR P Q R

EV0000261 · EV0000262 · EV0000263 · +24 source units in pack.json

16 Source slide deck: Problems of Lagranges equation

Study focus: Source slide deck: Problems of Lagranges equation. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000288 · EV0000289

SHORT-NOTE RECAP

Study notes · Source slide deck: Problems of Lagranges equation

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MATHEMATICS - I

- Unit - 3: Partial Differential Equations
- Department of Science and Humanities

Slide 2 · 1 Problems on Lagrange's Linear Equation

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · Solve p + xz q = y2

- This is Lagrange's equation Pp + Qq = R.
- The subsidiary equations are: dx = dy = dz P Q R
- Considering the first two ratios:

Slide 4 · Solve p + xz q = y2 (contd.)

- Z Z y3 x2dx = y2dy =⇒x3 = + c =⇒x3 −y3 = 3c
- Considering the first and last ratios:
- Z Z z2 x dx = z dz =⇒x2 = + c =⇒x2 −z2 = 2c =⇒x2 −z2 = b

Slide 5 · Solve yzp −xzq = xy

- From first and second fractions, we get
- x dx + y dy = 0
- From first and third fractions,

Slide 6 · Solve yzp −xzq = xy (contd.)

- Thus the general solution is
- ϕ(x2 + y2, x2 −z2) = 0

Slide 7 · Solve z(z2 + xy)(px −qy) = x4

- xz(z2 + xy) −yz(z2 + xy) x4
- From first and second fractions, we get

Slide 8 · Solve z(z2 + xy)(px −qy) = x4 (contd.)

- From first and third fractions,
- x4 −z4 −2(xy)z2 = c2
- ϕ(xy, x4 −z4 −2xyz2) = 0

EV0000288 · EV0000289 · EV0000290 · +21 source units in pack.json

17 Source slide deck: Method of separation of variables continued

Study focus: Source slide deck: Method of separation of variables continued. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000363 · EV0000364

SHORT-NOTE RECAP

Study notes · Source slide deck: Method of separation of variables continued

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MATHEMATICS - I

- Unit - 3: Partial Differential Equations
- Department of Science and Humanities

Slide 2 · 1 Method of Separation of Variables

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · Method of Separation of Variables

- The method of separation of variables is a powerful and
- Differential Equations (PDEs), especially when the PDE
- Separation of variables provides closed-form solutions for many fun

Slide 4 · Working Rule

- 1 For a P.D.E. in the function u of two independent
- variables x and y, assume that the required solution is
- 2 Substitution of u from (1) and its derivatives reduces the P.D.E. to

Slide 5 · Working Rule (contd.)

- 1 Thus (2) reduces to
- 2 The determination of a solution to P.D.E. reduces to the determi
- nation of solutions to two O.D.E.s (with appropriate conditions).

Slide 6 · Use the separation of variables technique to solve

- Solution: Assume u(x, y) = X(x)Y (y).
- Then the P.D.E. becomes 3X′Y + 2XY ′ = 0 or
- Given that 4e−x = u(x, 0) = X(x)Y (0) = cekx,

Slide 7 · Use separation of variables to solve

- 5 ux −4 uy = 0, u(x, 0) = 3e2x.
- Solution. Assume u(x, y) = X(x) Y (y). Then,
- u(x, y) = AB ekx e 4 k y = C ek (x+ 4 y).

EV0000363 · EV0000364 · EV0000365 · +18 source units in pack.json

18 Source slide deck: Particular integral of Type 1 and 2 Continued

Study focus: Source slide deck: Particular integral of Type 1 and 2 Continued. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000525 · EV0000526

SHORT-NOTE RECAP

Study notes · Source slide deck: Particular integral of Type 1 and 2 Continued

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MATHEMATICS - I

- Unit - 3: Partial Differential Equations
- Department of Science and Humanities

Slide 2 · 1 Problems when f(x, y) = eax+by, or

- f(x, y) = sin(ax + by) = cos(ax + by)

Slide 3 · Solve the partial differential equation:

- D2 + DD′ −2(D′)2 z = 5ex+2y
- To find CF: The factors are:
- Therefore, CF = ϕ1(2x −y) + ϕ2(−x −y) = ϕ1(2x −y) + ϕ2(x + y)

Slide 4 · Problem (contd.)

- D2 + DD′ −2(D′)2
- ⇒P.I. = 5 ex+2y
- General Solution is: z = ϕ1(2x −y) + ϕ2(x + y) −ex+2y

Slide 5 · Solve the partial differential equation

- (3D2 −2DD′ −D′2 + 4D′ −3D) z = 5 e(2x+y)/3
- To find CF: The factors are:
- Thus, CF is z = ϕ1(−x −y) + e−x 3 ϕ2(x −3y)

Slide 6 · Problem (contd.)

- F(D, D′) = 3D2 −2 DD′ −D′2 + 4D′ −3D.
- Since the RHS is 5e(2x+y)/3, we evaluate
- F 3, 3 = 3 3 −2 3 3 − 3 + 4 3 −3 3 = 9

Slide 7 · Problem (contd.)

- PI = · 5e(2x+y)/3 = · 5e(2x+y)/3 = 45e(2x+y)/3

Slide 8 · Solve the PDE

- (4D2 + 3DD′ −D′2 −D −D′)z = 3e(x+2y)/2
- Solution: To find DF: The factors are:
- For D + D′, a = b = 1. For 4D −D′ −1, a = 4, b = −1, c = −1.

Slide 9 · Problem (contd.)

- We write the particular integral as z = ϕ(x, y)e(x+2y)/2, where ϕ(x, y) is
- The given equation is: F(D, D′)[z] = 3e(x+2y)/2
- ⇒F D + 2,1 D′ + 1 · ϕ(x, y)e(x+2y)/2 = 3e(x+2y)/2

Slide 10 · Problem (contd.)

- F D + 2, D′ + 1 = 4 D + 2 + 3 D + 2 (D′ + 1) −(D′ + 1)2−
- = 6D −3 + 3DD′ + 4D2 −(D′)2 2D′
- = 6D 1 −1 + 1 + 4D −D′(D′)2} 4D′D−1 6{3D′

Slide 11 · Problem (contd.)

- ϕ(x, y) = F D + 2, D′ + 1 (3)
- ϕ(x, y) = 1 −1 + 1 + 4D −D′(D′)2} (1) 2D−1 4D′D−1 6{3D′
- Hence, the particular integral is given by

Slide 12 · Solve the PDE

- (2D2 + 5DD′ −3D′2) z = sin 2x −y .
- To find CF: The factors are
- For 2D −D′, a = 2, b = −1. For D + 3D′, a = 1, b = 3.

Slide 13 · Problem (contd.)

- We have the right hand side as sin(ax + by),
- F(D2, DD′, (D′)2) sin(2x −y) = [2(−4) + 5(2) −3(−1)] sin(2x −y)
- The particular integral is given by

EV0000525 · EV0000526 · EV0000527 · +36 source units in pack.json

19 Source slide deck: Particular integral of type 3 and 4 continued

Study focus: Source slide deck: Particular integral of type 3 and 4 continued. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000597 · EV0000598

SHORT-NOTE RECAP

Study notes · Source slide deck: Particular integral of type 3 and 4 continued

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MATHEMATICS - I

- Unit - 3: Partial Differential Equations
- Department of Science and Humanities

Slide 2 · 1 Problems on Case 3 and Case 4

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · Solve the PDE

- To find CF: The factors are
- For D + D′, a = b = 1. For D −2D′, a = 1, b = −2.
- Hence, the complementary function is

Slide 4 · Problem (contd.)

- We write 4D3−3D(D′)2+(D′)3 = 4D3h 1−34D−2(D′)2+ 4D−3(D′)3i1
- 1 3 i−1The PI is given by z = 4D−3h 1− 4D−2(D′)2−14D−3(D′)3 (6x2y2)
- Finally, the general solution is z = C.F + P.I

Slide 5 · Solve the PDE:

- To find CF: The factors are
- For (D + D′), a = b = 1. For 2D + D′ + 1, a = 2, b = 1, c = 1. Hence,
- z = ϕ1(x −y) + e 2 ϕ2(x −2y)

Slide 6 · Problem (contd.)

- The particular integral is given by
- i−1 z = D−1h 1 + {D−1D′ + 3D′ + 2D + D−1(D′)2} (x −y)
- Finally, the general solution is z = C.F + P.I

Slide 7 · Solve the PDE

- D2 −DD′ −2(D′)2 z = 16x e2y
- To find CF: The factors are
- Hence, the complementary function is

Slide 8 · Problem (contd.)

- The Particular integral is given by
- We first obtain (D + D′)−1(16xe2y).

- Denote u = (D + D′)−1(16xe2y), or (D + D′) u = 16xe2y

Slide 9 · Problem (contd.)

- The first two terms give y = x + c
- Using the first and third terms, we get
- Hence, u = 8 x −1 e2(x+c) = (8x −4)e2y 2

Slide 10 · Problem (contd.)

- Now, denote z = (D −D′)−1u = (D −D′)−1 (8x −4)e2y
- or (D −D′) z = (8x −4)e2y
- The first two terms give x + y = C

Slide 11 · Problem (contd.)

- Next, the first and third equations give
- ∴z = e2C (8x −4)e−2x dx
- ⇒z = e2C −4x e−2x = −4x e2C−2x = −4x e2(C−x) = −4x ey,

EV0000597 · EV0000598 · EV0000599 · +30 source units in pack.json

20 Source slide deck: Particular integral when X is any other form

Study focus: Source slide deck: Particular integral when X is any other form. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000630 · EV0000631

SHORT-NOTE RECAP

Study notes · Source slide deck: Particular integral when X is any other form

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MATHEMATICS - I

- Unit - 3: Partial Differential Equations
- Department of Science and Humanities

Slide 2 · 1 P.I. when f(x, y) = xmyn, or a polynomial in x, y

- 2 P.I. when f(x, y) is not of the form eax+by or sin(ax + b) or
- cos(ax + b) or xmyn (a polynomial in x, y)

Slide 3 · P.I. when f(x, y) = xmyn, or a polynomial in x, y

- We expand [F(D, D′)]−1 as an infinite series and operate on xmyn. If
- of D and D′, then we expand [F(D, D′)]−1 in powers of (D′)−1D if m < n
- and in powers of D−1D′ if m > n

Slide 4 · Find the particular integral of the partial differential equation

- Solution: We have [D2 −(D′)2]z = x2 + y2
- We write [D2 −(D′)2] = D2 1 −(D−1)2(D′)2
- ⇒[D2 −(D′)2]−1 = D−2 1 + (D−1)2(D′)2 + (D−1)4(D′)4 + . . .

Slide 5 · Problem (contd.)

- Therefore, the particular integral is given by
- PI = [D2 −(D′)2]−1(x2 + y2)
- In this case, we obtain a different form of the particular integral as

Slide 6 · Find the particular integral of the partial differential equation

- [D2 + 2DD′ + (D′)2]z = 3x + 2y, or [D + D′]2z = 3x + 2y
- [D + D′]2 = [D(1 + D′D−1)]2
- [D + D′]2 = D2(1 + D′D−1)2

Slide 7 · Problem (contd.)

- The particular integral is given by
- z = D−2(1 + D′D−1)−2(3x + 2y)
- = D−2[1 −2D′D−1 + 3(D′D−1)2 −. . .](3x + 2y)

Slide 8 · Problem (contd.)

- The particular integral is given by
- PI = (D′)−2[1 + (D′)−1D]−2(3x + 2y)
- = (D′)−2[3x −4y] On integrating twice w.r.t. y

Slide 9 · Let f(x, y) be not of any one of the forms as given in the

- consider one of the factors as
- Since this is a Lagrange's equation, the auxiliary equations are
- The first two ratios give the solution

Slide 10 · Consider now the first and third ratios:

- h i a1 dz = f x, (b1x −c)/a1 dx.
- where c1 is an arbitrary constant. After integration, we replace c by
- b1x −a1y as given in Eq. 1. Since a particular integral is required, we

Slide 11 · z = F x, b1x −a1y

- We repeat the procedure for each factor to obtain the required

EV0000630 · EV0000631 · EV0000632 · +30 source units in pack.json

21 Source slide deck: Particular integral when X is any other form continued

Study focus: Source slide deck: Particular integral when X is any other form continued. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000663 · EV0000664

SHORT-NOTE RECAP

Study notes · Source slide deck: Particular integral when X is any other form continued

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MATHEMATICS - I

- Unit - 3: Partial Differential Equations
- Department of Science and Humanities

Slide 2 · 1 Problems on special case

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · Find the solution of the differential equation

- The particular integral is given by
- We first obtain (D + D′)−1(yex) as in case 4. For the sake of
- completeness, we repeat the procedure used in this case. Denote

Slide 4 · Problem (contd.)

- The first two terms give y = x + c. Using the first and third terms, we
- u = (x + c)exdx = (x + c −1)ex = (y −1)ex.
- z = (2D + 3D′)−1u = (2D + 3D′)−1(y −1)ex

Slide 5 · Problem (contd.)

- The first two terms give 2y = 3x + c1. The first and third terms give
- z = (3x + c1 −2)exdx = + c1 −5)ex = −5)ex 4 4(3x 4(2y
- z = ϕ1(3x −2y) + ϕ2(x −y) + 4(2y −5)ex.

Slide 6 · Find the solution of the differential equation

- D2 + D D′ −2(D′)2 z = 8 ln x + 5y
- We write D2 + D D′ −2(D′)2 z = (D + 2D′)(D −D′) z = 8 ln x + 5y .

- z = ϕ1(2x −y) + ϕ2(x + y)

Slide 7 · Problem (contd.)

- The particular integral is given by
- z = (D + 2D′)−1 (D −D′)−1 8 ln(x + 5y)
- u = (D −D′)−1 8 ln(x + 5y) , or (D −D′) u = 8 ln(x + 5y).

Slide 8 · Problem (contd.)

- The first and second terms give x + y = c.
- The first and third terms give =
- u = 8 ln 5c −4x dx = 8 x ln 5c −4x + dx

Slide 9 · Problem (contd.)

- ⇒u = 8 ln 5c −4x dx = 8 x ln 5c −4x − dx
- ⇒u = 8 ln 5c −4x dx = 8 x ln 5c −4x − 1 + dx
- ⇒u = 8 x ln 5c −4x −x − dx

Slide 10 · Problem (contd.)

- ⇒u = 8 x ln 5c −4x −x −5c ln(5c −4x)
- h ⇒u = 8 x −5c ln 5c −4x −xi 4
- ⇒u = −2(x + 5y) ln(x + 5y) −8x

Slide 11 · Problem (contd.)

- Now, denote z = (D + 2D′)−1u
- or (D + 2D′) z = u = − 8x + 2(x + 5y) ln(x + 5y)
- The first two terms give 2x −y = c1

Slide 12 · Problem (contd.)

- Integrate w.r.t. x by taking 11x −5c1 = t. Then
- 2 1 1 i h z = − 4x2 + 2 (11x −5c1)2 ln(11x −5c1) − 4 (11x −5c1)2 11
- 1 1 h ⇒z = − 4x2 + (11x −5c1)2 ln(11x −5c1) − 22 (11x −5c1)2i 11

Slide 13 · Problem (contd.)

- Since 2x −y = c1, 5c1 = 10x −5y. Then 11x −5c1 becomes x + 5y.
- The general solution of the differential equation is
- z = ϕ1(2x−y) + ϕ2(x+y) − 4x2 + 22(x+ 5y)2 2 ln(x+5y) −1

EV0000663 · EV0000664 · EV0000665 · +36 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Formation of PDE transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Formation of PDE transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Formation of PDE transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Formation of PDE transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Introduction transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Introduction transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on formation of PDE by eliminating arbitrary functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on formation of PDE by eliminating arbitrary functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on formation of PDE by eliminating arbitrary functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Problems on formation of PDE by eliminating arbitrary functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Lagranges Linear equation_continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Lagranges Linear equation_continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Modeling of One dimensional Heat equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Modeling of One dimensional Heat equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Modeling of One dimensional Heat equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9974; only the JSON segmentation supplies the displayed timestamps.
Modeling of One dimensional Heat equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Solution of one dimensional heat equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Solution of one dimensional heat equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Solution of one dimensional heat equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Solution of one dimensional heat equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of type 1 and 2 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of type 1 and 2 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of type 1 and 2 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Particular integral of type 1 and 2 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on finding the solution when RHS=0 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on finding the solution when RHS=0 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on finding the solution when RHS=0 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9965; only the JSON segmentation supplies the displayed timestamps.
Problems on finding the solution when RHS=0 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral -Type 3 and Type 4 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Formation of PDE	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Formation of PDE	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Formation of PDE (converted PDF; original hash 6b7579603cfb0efd48bd28bfbf4ef0e5efc3f852b845e4c5d58726437cc5d251)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Introduction to Partial differentiation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Introduction to Partial differentiation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Introduction to Partial differentiation (converted PDF; original hash 496c28b2e6072e7dd4a2fa45951cc9287e90f3476ebfad3c06ff26caad113eba)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems on formation of PDE by eliminating arbitrary functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Problems on formation of PDE by eliminating arbitrary functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Problems on formation of PDE by eliminating arbitrary functions (converted PDF; original hash 6088fd4554e5221d7231bb022216ee5422c61c5a8b21fea2359fb7442ae4649d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Lagrange s Linear Equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Lagrange s Linear Equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Lagrange s Linear Equation (converted PDF; original hash 0f0d47015a775c00d5a2ef1c1ba0ad54770829f306f718a0ec8b0ae8aa28f548)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems of Lagranges equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Problems of Lagranges equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Problems of Lagranges equation (converted PDF; original hash 45fa62ce35f0b306eab5a3f9dadf64d5e48dbbca35e38a017405d89268100242)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Lagrange s Linear Equation Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Lagrange s Linear Equation Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Lagrange s Linear Equation Continued (converted PDF; original hash 1629fb7dfc2d5895a175d48266cf8dcad23def6e219976b01adf7120e45ef814)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Method of separation of variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Method of separation of variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Method of separation of variables (converted PDF; original hash 78c0cbeb9322daba54ecea6092cb185fa60b5fa52d896ac35042c6c3e27988b7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Method of separation of variables continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Method of separation of variables continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Method of separation of variables continued (converted PDF; original hash 22c930a9df5e782fd0e32a2cc9faf75db1ff5cb668310cbd24d0412b87da02da)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Modeling of one dimensional heat equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Modeling of one dimensional heat equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Modeling of one dimensional heat equation (converted PDF; original hash bf2d6d6926b3ddb1f00f641cefc01736225035f2f8799c95ed0da06bac5074a6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Solution of one dimensional heat equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Solution of one dimensional heat equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Solution of one dimensional heat equation (converted PDF; original hash ae4bcbac05a2c15a9a250781a63db204cafb89f00bdb705917d9af11397ae05b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Higher order PDE	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Higher order PDE	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Higher order PDE (converted PDF; original hash 2b96649655e0e57d5e1312ee2ae72dcc1948ceca02bd1d6edf53d0f772ee6a5d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of Type 1 and 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Particular integral of Type 1 and 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of Type 1 and 2 (converted PDF; original hash 3d506acd6683b2c621e0d15c9ca59f1065600dc4cc87198b47d515e6e31afee0)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: solutions of PDE when RHS 0	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: solutions of PDE when RHS 0	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for solutions of PDE when RHS 0 (converted PDF; original hash ce9d6509f0e7ebe5773e99c5c9758150bcc91a4cc9d770d5d396e9dd59b58680)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of Type 1 and 2 Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Particular integral of Type 1 and 2 Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of Type 1 and 2 Continued (converted PDF; original hash 0675c88ff179ac22172f95bc9add1e205ebd6cc4ef187b2ed1404995555316e5)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of type 3 and 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral of type 3 and 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of type 3 and 4 (converted PDF; original hash a16d85b951f83cc4d5d75cfe114ee11333442d8e6eb035d99e177fbafcd65086)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of type 3 and 4 continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Particular integral of type 3 and 4 continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of type 3 and 4 continued (converted PDF; original hash 3bcb225c312ddc18cf5f74b2866c07d37c692b6ee7dd25b47d3355eda33f8359)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral when X is any other form	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral when X is any other form	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral when X is any other form (converted PDF; original hash a16d85b951f83cc4d5d75cfe114ee11333442d8e6eb035d99e177fbafcd65086)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral when X is any other form continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Particular integral when X is any other form continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral when X is any other form continued (converted PDF; original hash 6ff6d8dbbbe10faa89598152440e66fd69fddf04c79cd701298c561fa443f4c4)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare...
SRC00003	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare...
SRC00005	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00006	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare...
SRC00007	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare...
SRC00008	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare...
SRC00010	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare...
SRC00012	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare...
SRC00014	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare...
SRC00016	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare...
SRC00018	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00019	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare...
SRC00020	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00021	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare...
SRC00022	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00023	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...
SRC00024	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00025	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...
SRC00026	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00027	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...
SRC00028	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...
SRC00030	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00031	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare...
SRC00032	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00033	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare...
SRC00034	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00035	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00036	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00037	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00038	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...
SRC00040	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...
SRC00042	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00043	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare...
SRC00044	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00045	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare...
SRC00046	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare...
SRC00047	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00048	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare...
SRC00049	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare...
SRC00050	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00051	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare...
SRC00052	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00053	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare...
SRC00054	0 covered · 0 excluded · 11 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00056	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00055	0 covered · 0 excluded · 11 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00057	0 covered · 0 excluded · 10 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00059	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00058	0 covered · 0 excluded · 10 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00060	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00062	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00061	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00063	9 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00065	9 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00064	9 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00066	8 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00068	8 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00067	8 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00069	0 covered · 0 excluded · 10 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00071	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00070	0 covered · 0 excluded · 10 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00072	0 covered · 0 excluded · 7 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00074	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00073	0 covered · 0 excluded · 7 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00075	7 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00077	7 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00076	7 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00078	0 covered · 0 excluded · 7 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00080	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00079	0 covered · 0 excluded · 7 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00081	0 covered · 0 excluded · 6 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00083	0 covered · 0 excluded · 6 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00082	0 covered · 0 excluded · 6 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00084	0 covered · 0 excluded · 16 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00086	0 covered · 0 excluded · 16 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00085	0 covered · 0 excluded · 16 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00087	0 covered · 0 excluded · 11 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00089	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00088	0 covered · 0 excluded · 11 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00090	0 covered · 0 excluded · 7 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00092	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00091	0 covered · 0 excluded · 7 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00093	13 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00095	13 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00094	13 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00096	0 covered · 0 excluded · 11 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00098	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00097	0 covered · 0 excluded · 11 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00099	11 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00101	11 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00100	11 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00102	11 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00104	11 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00103	11 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00105	13 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00107	13 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00106	13 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

C Revision and retrieval practice